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Research interests

Environmental Epidemiology; Air pollution; Lung Health; Environmental Hazard and Cancer Development

Education Background

- B.S., Health Inspection and Quarantine, Chongqing Medical University. Chongqing, China
(09/2013—06/2017)
- M.Med., Health Inspection and Quarantine, Chongqing Medical University. Chongqing, China
(09/2017—06/2020)
- PhD, Environmental Epidemiology, The University of Hong Kong. Hong Kong, China
(09/2022—Current)

Publications

- **Ji Renyue**, Wu, H., Lin, H., Li, Y., & Shi, Y. Cadmium and selenium blood levels in association with congestive heart failure in diabetic and prediabetic patients: a cross-sectional study from the national health and nutrition examination survey. *Diabetology & Metabolic Syndrome*; 2025, 17(1), 12.
- Li, J., Wu, H., Xing, W., Li, X., Han, Z., **Ji Renyue**, Deng, Z., Jung, M., Sun, S., Chung, B.I., Cardenas, A., Langston, M.E. Air pollution mixture associated with oxidative stress exacerbation and symptoms deterioration in allergic rhinitis patients: Evidence from a panel study. *Science of The Total Environment*; 2024, 172688
- Zhang, J., He, Q., Du, L., **Ji Renyue**, Yuan, L., Zhang, H., ... & Li, Y. Risk factor for lung infection in recipients after liver transplantation: A meta-analysis. *Artificial Organs*; 2021, 45(3), 289-296.
- **Ji Renyue**, Niu, W., Chen, S., Xu, W., Ji, X., Yuan, L., Zhao, H., Geng, M., Qiu, J., Li, C. Target-inspired Pb²⁺-dependent DNAzyme for ultrasensitive electrochemical sensor based on MoS₂-AuPt nanocomposites and hemin/G-quadruplex DNAzyme as signal amplifier. *Biosensors and Bioelectronics*; 2019, 144: 111560.
- **Ji Renyue**, Chen, S., Xu, W., Qin, Z., Qiu, J., Li, C. A voltammetric immunosensor for clenbuterol based on the use of a MoS₂-AuPt nanocomposite. *Microchimica Acta*; 185 (2018) 209

Research Project

The Xuan Wei area of Yunnan Province, in the southwest of China, bears the abnormally high incidence and mortality rate of lung cancer in the world. Specifically, the mortality rate of females, all of whom are non-smokers, is the highest worldwide. Epidemiological studies have suggested that this has been associated with the domestic combustion of C1 coal from high risk area of lung cancer in Xuan Wei, which contains iron-rich chamosite. Therefore, my research project is to explore the underlying molecular mechanism of lung cancer induced by chamosite-containing C1 coal. My study will provide experimental insights into how particles in C1 coal contribute to the development and progression of lung cancer, which may provide potential systemic interventions and measurements on Xuan Wei lung cancer prevention and control.

Research Skills

- Computer Skills: R Language, SPSS, Endnote, ImageJ, GraphPad Prism, AdobePhotoshop
- Bioinformatics Skills: Genomic data analysis (single cell-RNA-seq, Bulk-RNA-seq)
- Biological experimental skills: DNA/RNA/protein extraction and purification, q-PCR, RT-PCR, molecular cloning, cell culture, cell transfection, cell proliferation assays , flow cytometry, immunofluorescence staining, HE staining, cell migration and invasion assays, western blotting, ELISA, animal model establishment, animal surgery (injections, dissection, tissue sampling), light /confocal/electron microscopy (TEM, SEM)

Awards & honors

- HKU Postgraduate Scholarship, The University of Hong Kong, 2022.9 - current
- Outstanding master's degree paper prize, Chongqing Medical University, 2020.1
- Advanced individual award for scientific and technological innovation, Chongqing Medical University, 2020.1
- **National Scholarship, Ministry of Education of China, 2019.12**
- Scholarship for Postgraduate, Chongqing Medical University, 2019.6
- **National Scholarship, Ministry of Education of China, 2018.11**
- Scholarship for Postgraduate, Chongqing Medical University, 2018.6
- Honor of the “Outstanding Graduates”, Chongqing Medical University, 2017.4
- Honor of the “Excellent Student Cadre”, Chongqing Medical University, 2016.10
- Honor of the “Outstanding Volunteer”, Chongqing Medical University, 2016.6